

556 Mar 18, 2026

556.1 Review

When you compose two transducers the resulting transducer's linear operator is the tensor product of its components. Consider for instance an input-agnostic transducer (HMM), that outputs the X process with linear operator in the $\mathcal{R}$ space

$$\hat{T}^{(x)} = \sum_{r, r'} T_{r \rightarrow r'}^{(x)} |r'\rangle \langle r|, \quad (6734)$$

and a transducer that transforms x to y in the $\mathcal{S}$:

$$\hat{U}^{(y|x)} = \sum_{s, s'} U_{s \rightarrow s'}^{(y|x)} |s'\rangle \langle s|. \quad (6735)$$

The linear operator for the resulting HMM that produces the joint process is

$$\hat{V}^{(xy)} = \hat{U}^{(y|x)} \otimes \hat{T}^{(x)}, \quad (6736)$$

which can be used to calculate probabilities

$$\Pr(Y_{0:L} = y_{0:L}, X_{0:L} = x_{0:L}) = \langle 1|_R \otimes \langle 1|_S \prod_{t=0}^{L-1} \hat{V}^{(x_t y_t)} | \rho_{R_0 \otimes S_0} \rangle. \quad (6737)$$

This means that the belief states of joint process approach tensor product states. Furthermore, this implies potential dimensional advantages: that the latents of model can be linearly expressed in $(|\mathcal{R}| - 1) + (|\mathcal{S}| - 1)$ dimensions rather than the usual $|\mathcal{R}||\mathcal{S}| - 1$ dimensions for a simplex in the joint space.

556.2 Complexifying by coarse-graining observables

While ignoring the output y necessarily simplifies the network, ignoring x can actually add complexity to the process. This can be seen by constructing the HMM operator $\hat{W}^{(y)}$ for the Y process and examining the dimension of the belief states. To access the HMM for the Y process, simply coarse-grain the joint process

$$\Pr(Y_{0:L} = y_{0:L}) = \sum_{x_{0:L}} \langle 1|_R \otimes \langle 1|_S \prod_{t=0}^{L-1} \hat{U}^{(y_t|x_t)} \otimes \hat{T}^{(x_t)} | \rho_{R_0 \otimes S_0} \rangle \quad (6738)$$

$$= \langle 1|_R \otimes \langle 1|_S \prod_{t=0}^{L-1} \left(\sum_{x_t} \hat{U}^{(y_t|x_t)} \otimes \hat{T}^{(x_t)} \right) | \rho_{R_0 \otimes S_0} \rangle \quad (6739)$$

$$= \langle 1|_R \otimes \langle 1|_S \prod_{t=0}^{L-1} \hat{W}^{(y_t)} | \rho_{R_0 \otimes S_0} \rangle, \quad (6740)$$

where

$$\hat{W}^{(y)} = \sum_x \hat{U}^{(y|x)} \otimes \hat{T}^{(x)}. \quad (6741)$$

Because of the sum over x , the $\mathcal{S}$ and $\mathcal{R}$ subspaces are not necessarily separable into a tensor product.

In fact, the full variety of behavior belief-state geometries in the $|\mathcal{R}||\mathcal{S}|$ -simplex is accessible through such operators. This can be seen by considering the following example for generating any HMM over the joint space.

556.3 Example of complexification

Consider an arbitrary HMM that produces a process over Y in the joint latent space $R \otimes S$:

$$M_{rs \rightarrow r's'}^{(y)} = \Pr(R_{t+1} = r', S_{t+1} = s', Y_t = y | R_t = r, S_t = s). \quad (6742)$$

Let $\mathcal{X}$ be equal to the size of the joint memory space squared, meaning that it realizes elements of the form $rr'ss'$. You can think of $\mathcal{X}$ as encoding the elements of a matrix in the joint space. In this case, we can choose

$$T_{r \rightarrow r'}^{(x)} = \Pr(R_{t+1} = r' | R_t = r, S_t = s) \delta_{rr'ss'=x} \quad (6743)$$

$$U_{s \rightarrow s'}^{(y|x)} = \sum_{r, r'} \Pr(S_{t+1} = s', Y_t = y | R_{t+1} = r', R_t = r, S_t = s) \delta_{rr' \in x}, \quad (6744)$$

where $\delta_{rr' \in x}$ evaluates to 1 if ss' are part of the sequence x and 0 otherwise. The result of applying the transducer to the HMM and coarse-graining the input yields the expected joint HMM

$$\sum_x U_{s \rightarrow s'}^{(y|x)} \otimes T_{r \rightarrow r'}^{(x)} = \sum_x \Pr(R_{t+1} = r' | R_t = r, S_t = s) \delta_{rr'ss'=x} \quad (6745)$$

$$\times \sum_{r'', r'''} \Pr(S_{t+1} = s', Y_t = y | R_{t+1} = r''', R_t = r'', S_t = s) \delta_{r''r''' \in x} \quad (6746)$$

$$= \Pr(R_{t+1} = r' | R_t = r, S_t = s) \quad (6747)$$

$$\times \sum_{r'', r'''} \Pr(S_{t+1} = s', Y_t = y | R_{t+1} = r''', R_t = r'', S_t = s) \delta_{r''r''' = rr'} \quad (6748)$$

$$= \Pr(R_{t+1} = r' | R_t = r, S_t = s) \Pr(S_{t+1} = s', Y_t = y | R_{t+1} = r', R_t = r, S_t = s) \quad (6749)$$

$$= \Pr(R_{t+1} = r', S_{t+1} = s', Y_t = y | R_t = r, S_t = s) \quad (6750)$$

$$= M_{rs \rightarrow r's'}^{(y)}. \quad (6751)$$

To confirm that these are valid transducers/HMMs, we can calculate the normalization conditions:

$$\sum_{x,r'} T_{r \rightarrow r'}^{(x)} = \sum_{x,r'} \Pr(R_{t+1} = r' | R_t = r, S_t = s) \delta_{rr' ss' = x} \quad (6752)$$

$$= \sum_{x,r'} \Pr(R_{t+1} = r' | R_t = r, S_t = s) \delta_{rr' ss' = x} \quad (6753)$$

$$= \sum_{r'} \Pr(R_{t+1} = r' | R_t = r, S_t = s) \quad (6754)$$

$$= 1 \quad (6755)$$

$$T \text{ is normalized} \quad (6756)$$

$$\sum_{s',y} U_{s \rightarrow s'}^{(y|x)} = \sum_{s',y} \sum_{r,r'} \Pr(S_{t+1} = s', Y_t = y | R_{t+1} = r', R_t = r, S_t = s) \delta_{rr' \in x} \quad (6757)$$

$$= \sum_{r,r'} \delta_{rr' \in x} \quad (6758)$$

$$= 1 \quad (6759)$$

$$U \text{ is normalized.} \quad (6760)$$

Thus, we have a way of generating any HMM over the joint space $R \otimes S$ that produces a Y process, meaning that we can access any belief geometry of that joint space by coarse-graining a high-dimensional input variable.

Questions

1. How does the size of x affect the dimension and extent of the belief geometry?
2. When do we see complexification vs. simplification?
3. What are principles for designing HMMs that span the space of the simplex?

556.4 Example of exponential complexification

We can observe a very large increase in dimension if the output transducer is a collection of different parallel transducers

$$\vec{\hat{U}}^{(\vec{y}|\vec{x})} = \bigotimes_{n=0}^{N-1} \hat{U}^{(y(n)|x(n))}_n, \quad (6761)$$

where $x(n)$ is the input to agent n , $y(n)$ is its output, and $S(n)$ is its internal state space. This basically simulates a hierarchy where a central process T dictates the behavior of N independent agents $\hat{U}(n)$. The resulting object can be

efficiently modeled with $(|\mathcal{R}|0 - 1) + \sum_{n=0}^{N-1} (|\mathcal{S}(n)|0 - 1)$ dimensions. However if we coarse-grain the inputs $x(n)$, losing track of the orders driving each agent, suddenly the potential state-space explodes to $|\mathcal{R}| \prod_{n=0}^{N-1} |\mathcal{S}(n)| - 1$. The result is potentially exponentially more complex, in terms of the number of dimension necessary to predictively model the system.

There are a few interesting take-aways from this... For instance, this is why centralized conspiracies can be so confusing. When agents are all taking orders from a central process, but that process is hidden, it can be extremely challenging to efficiently model observations.